

CANALYTICS

Suez Canal Shipments Analysis

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- The Suez Canal is one of the most strategic maritime routes globally
- Connects Europe, Asia, and the Middle East
- Handles a significant share of global trade
- Operational disruptions or inefficiencies have global economic impact



Business Problem & Objectives

- Lack of centralized visibility on:
  - Traffic volume
  - Revenue and costs
  - Delays and operational efficiency
- Project objectives:
  - Monitor canal performance
  - Identify operational bottlenecks
  - Analyze revenue and profitability drivers
  - Support strategic and financial decision-making

Analytical Questions

- Which vessel types generate the most traffic?
- Which cargo categories are most profitable?
- How do delays vary by vessel type?
- Which regions contribute most to revenue?
- How does revenue evolve Month-over-Month and Year-over-Year?



Dataset Overview

- Dataset: “suez_canal_shipments.csv”
- Timeframe: 2015–2024
- Size: ~10,000+ shipment records
- Grain: 1 row = 1 shipment transit
- Key fields:
  - Shipment date
  - Vessel type
  - Cargo type
  - Origin & destination regions
  - Revenue, cost, delay, cargo volume

Data Preparation (ETL)

- Data cleaning and formatting
- Standardization of vessel and cargo categories
- Creation of derived fields (Month, Year, Month-Year)
- Validation of numerical fields (revenue, cost, delays)
- Preparation of analysis-ready dataset

```
text_cols = df.select_dtypes(include="object").columns

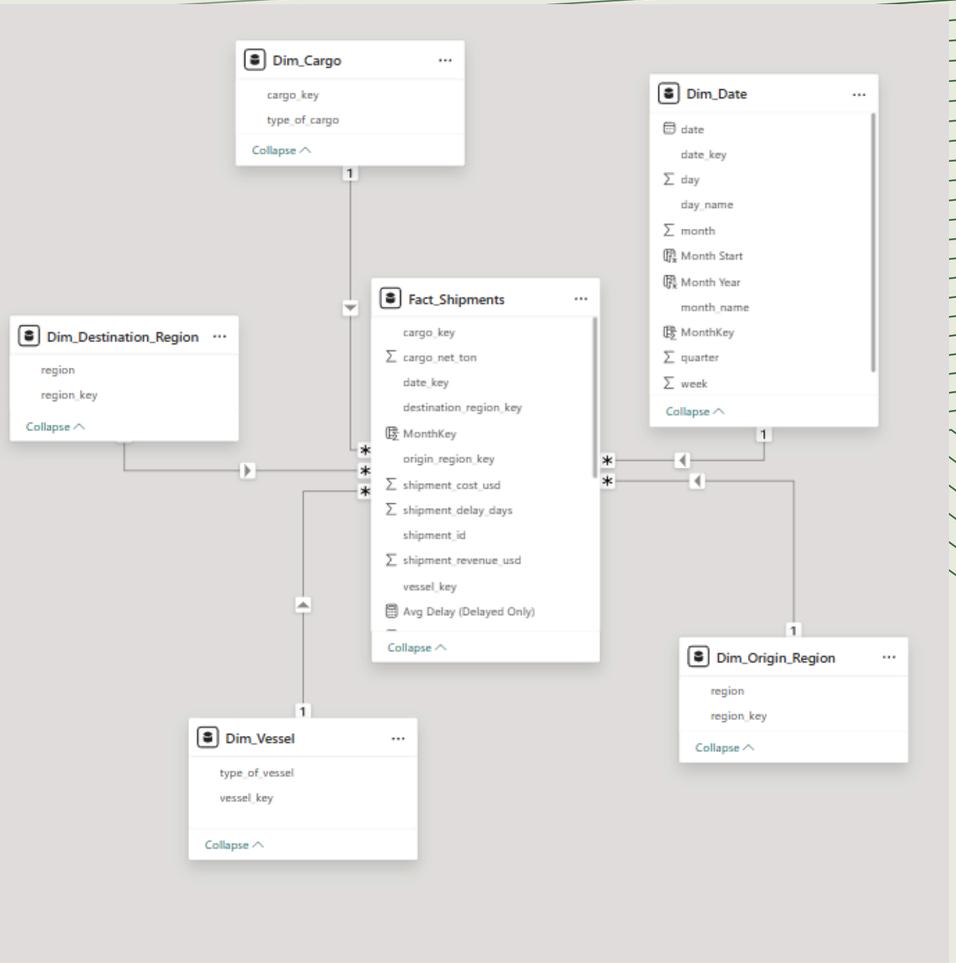
for col in text_cols:
    df[col] = (
        df[col]
        .astype(str)
        .str.strip()
        .str.replace(r"[\n\t]", "", regex=True)
        .str.replace(r"\*+|\?+", "", regex=True)
        .replace("nan", np.nan)
    )
```

```
num_cols = df.select_dtypes(include=["int64", "float64"]).columns
cat_cols = df.select_dtypes(include="object").columns

df[num_cols] = df[num_cols].fillna(df[num_cols].median())
df[cat_cols] = df[cat_cols].fillna("Unknown")
```

Data Modeling (Star Schema)

- Dimensional modeling approach
- Star schema design:
  - Fact table: Shipments
- Dimensions:
  - Date
  - Vessel
  - Cargo
  - Origin Region
  - Destination Region
- Benefits:
  - High performance
  - Easy analysis in Power BI
  - Clear business logic



Key KPIs & Measures

- **Operational KPIs:**
  - Number of transits
  - Average delay
  - Delay rate
- **Financial KPIs:**
  - Total revenue
  - Total cost
  - Profit
  - Revenue MoM (% change)
  - Revenue YoY (% change)

```
Revenue YoY =
VAR LY = [Revenue Last Year]
RETURN
IF (
    ISBLANK ( LY ) || LY = 0,
    BLANK (),
    DIVIDE ( [Total Revenue (USD)] - LY, LY )
)
```

```
Revenue MoM =
IF (
    ISINSCOPE ( Dim_Date[month] ) || ISINSCOPE ( Dim_Date[month_name] ),
    [Revenue MoM (Monthly)],
    VAR CY = MAX ( Dim_Date[year] )
    VAR FirstM =
        CALCULATE ( MIN ( Dim_Date[month] ), FILTER ( ALL ( Dim_Date ), Dim_Date[year] = CY ) )
    VAR LastM =
        CALCULATE ( MAX ( Dim_Date[month] ), FILTER ( ALL ( Dim_Date ), Dim_Date[year] = CY ) )
    VAR RevFirst =
        CALCULATE ( [Total Revenue (USD)], FILTER ( ALL ( Dim_Date ), Dim_Date[year] = CY && Dim_Date[month] = FirstM ) )
    VAR RevLast =
        CALCULATE ( [Total Revenue (USD)], FILTER ( ALL ( Dim_Date ), Dim_Date[year] = CY && Dim_Date[month] = LastM ) )
    RETURN
        IF ( ISBLANK ( RevFirst ) || RevFirst = 0, BLANK(),
            DIVIDE ( RevLast - RevFirst, RevFirst )
        )
)

```

Power BI Report



Key Findings

Total Revenue
$24.91B

Total Profit
$5.66B

Profit Margin
~22.7%

Delay Rate
~88%

On-Time Transits
1,338

Top Cargo
Grain – $4.31B

Key Findings

- **Container Ships are the core engine**
  - 57% of profit ($3.24B) | Lowest delay (4.63 days)
- **Delays are the #1 operational risk**
  - 88% of transits delayed (10,010 / 11,348)
  - LNG & Tankers have the longest delays (5.3–5.4 days)
- **Profit is highly concentrated**
  - Container + Tanker vessels = ~86% of total profit
  - LNG = <1% profit
- **Middle East leads revenue**
  - 21.1% of total revenue (highest regional share)
- **Revenue volatility is rising**
  - 2024 YoY revenue: –34.9%
  - Indicates demand or cost pressure

Business Recommendations

- Reduce transit delays (highest priority)
- Protect container ship traffic (57% of profit)
- Optimize tanker operations
- Improve scheduling and slot allocation
- Focus on efficiency, not volume

Future Enhancements

- Real-time vessel tracking
- Weather and incident data integration
- Predictive congestion analytics
- Port-level impact analysis (Rades, Sfax)
- Freight cost and fuel data integration
- Emissions and sustainability metrics



Thank you.